Quantum Mechanics II — Exercise Sheet 9

Weyl operators and phase space

Variant: Solutions

Exercise 1 (A closed displacement path). Use the convention $W(q,p) = e^{-\mathrm{i}qp/(2\hbar)}V(p)U(q)$. (i) Compute $W(q,0)W(0,p)W(-q,0)W(0,-p)$. Check the result by acting on a Schwartz function. (ii) Explain which phase is observable if this path is compared coherently with doing nothing.

Solution. The product is $U(q)V(p)U(-q)V(-p)$. Using $U(q)V(p) = e^{-\mathrm{i}qp/\hbar}V(p)U(q)$ gives

$$W(q,0)W(0,p)W(-q,0)W(0,-p) = e^{-\mathrm{i}qp/\hbar}\mathbf{1}.$$

Acting from right to left on $f(x)$ produces the factors $e^{-\mathrm{i}px/\hbar}$, then $e^{-\mathrm{i}p(x+q)/\hbar}$, then $e^{-\mathrm{i}qp/\hbar}$, and finally the same constant after the last translation. The single output vector differs only by a global phase. In a coherent comparison of the two paths the relative phase is $-qp/\hbar$ modulo 2π.

Exercise 2 (A reducible Weyl pair). On $L^2(\mathbb{R})\otimes\mathbb{C}^2$ let $U(a) = U_{\mathrm{line}}(a)\otimes\mathbf{1}$ and $V(b) = V_{\mathrm{line}}(b)\otimes\mathbf{1}$. Check Weyl's relation and strong continuity. Give a nontrivial closed subspace invariant under both groups and a nonscalar bounded operator commuting with them. Explain why the irreducible Stone–von Neumann theorem does not apply.

Solution. Write a vector as a pair (f_1, f_2). Both groups act by the line operator on each component, so Weyl's relation holds on both components with the same phase. The sum of the two squared norms proves strong continuity from that of the line groups. The subspace $L^2(\mathbb{R})\otimes\mathrm{span}\{(1,0)\}$ is nonzero, closed, proper and invariant. The projection onto it is

$$\Pi = \mathbf{1}\otimes\begin{pmatrix}1 & 0\\ 0 & 0\end{pmatrix}.$$

It commutes with both groups and is not scalar. The required irreducibility hypothesis fails.

Exercise 3 (The Wigner function of a Gaussian). For $s>0$ let $\psi(x) = (\pi s^2)^{-1/4}e^{-x^2/(2s^2)}$. Define its Wigner function by

$$F_\psi(q,p) = \frac{1}{2\pi\hbar}\int_{\mathbb{R}} e^{-\mathrm{i}py/\hbar}\psi(q+y/2)\overline{\psi(q-y/2)}\,\mathrm{d}y.$$

Compute F_ψ and its two marginals. Use the Gaussian integral

$$\int e^{-ay^2-\mathrm{i}by}\,\mathrm{d}y = \sqrt{\pi/a}\,e^{-b^2/(4a)}$$

for $a>0$. Does positivity for this one state establish a joint position and momentum observable?

Solution. Multiplying the two shifted Gaussians gives the exponent $-q^2/s^2 - y^2/(4s^2)$. The given integral yields

$$F_\psi(q,p) = \frac{1}{\pi\hbar} e^{-q^2/s^2 - s^2p^2/\hbar^2}.$$

Integrating first over p or over q gives respectively

$$\int F_\psi(q,p)\,\mathrm{d}p = \frac{e^{-q^2/s^2}}{\sqrt{\pi}s}, \qquad \int F_\psi(q,p)\,\mathrm{d}q = \frac{s}{\sqrt{\pi}\hbar} e^{-s^2p^2/\hbar^2}.$$

These are the position and momentum densities of this state. Positivity here is a property of one function, not a construction of a joint spectral measure for the noncommuting observables Q and P.

Exercise 4 (A displacement through one oscillator period). Let H be the oscillator Hamiltonian, $U_t = e^{-\mathrm{i}tH/\hbar}$, and let $W(q,p)$ act on the oscillator ground state. Using its coherent-state interpretation, find $\langle Q\rangle_t$ and $\langle P\rangle_t$. Determine $U_{2\pi/\nu}W(q,p)h_0$ as a vector, not only as a physical state.

Solution. The displacement puts the ground-state means at q and p. The coherent amplitude rotates by $e^{-\mathrm{i}\nu t}$, so

$$\langle Q\rangle_t = q\cos\nu t + \frac{p}{m\nu}\sin\nu t, \qquad \langle P\rangle_t = p\cos\nu t - m\nu q\sin\nu t.$$

Every number eigenvector acquires the phase $e^{-2\pi\mathrm{i}(n+1/2)} = -1$ after a full period. Consequently

$$U_{2\pi/\nu}W(q,p)h_0 = -W(q,p)h_0.$$

The means and the physical state return, while the vector changes sign.